

DANIEL SZWARC

AI Automation Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

AI Automation Engineer and Full-Stack Developer with 20+ years of enterprise application development experience, completing an MSc in Artificial Intelligence at the University of Liverpool (expected 2026). Specialized in agentic workflows, LLM orchestration, RAG-based applications, and end-to-end product ownership. Proven track record delivering production AI systems across healthcare, manufacturing, and financial services. Comfortable operating across the full stack, from system architecture and API design to frontend and deployment.

CURRENT AI PROJECTS

BeyondTheProtocol | *Production Clinical AI Platform*

- Built a private medical intelligence platform for a patient with ultra-rare metastatic cancer, processing 289+ anonymized clinical documents into a unified dashboard with lab trend charts, clinical alerts, and treatment timeline
- RAG-powered AI assistant (Claude + pgvector) answers clinical questions in EN/ES with mandatory source citations, 4-layer prompt injection defense, full audit log, and RBAC
- Architected and deployed end to end: full bilingual support (EN/ES with DB-stored preferences), mobile-responsive UI, and automated anonymization pipeline
- Stack: Next.js 15, FastAPI, PostgreSQL/pgvector, Docker/Traefik, Anthropic API, OpenAI embeddings, self-managed VPS

WorkflowSynth | *MSc Capstone Research Project*

- Designing a system that converts natural language descriptions into structured, formally verified executable workflows (n8n / LangChain) using a two-layer DSL and LLM-guided iterative refinement
- Architecture: LangGraph state machine, Lark-based formal grammar, 25 constrained DSL primitives, verify-then-repair loop (max 10 attempts), two-stage DSL and output verification
- Stack: Python, LangChain, LangGraph, Lark; output targets: n8n JSON and LangChain Python

APiWiki | *SOX-Compliant Internal Knowledge Base | Fortune 500 Client*

- Architected a RAG-powered internal knowledge platform with SOX-compliant on-premises LLM inference, Markdown-to-HTML pipeline, audit log UI with RBAC, and injection-defended chat interface
- Stack: Next.js, FastAPI, PostgreSQL/pgvector, Docker, local Ollama inference

MedSynth | *Healthcare AI Platform*

- Built a full-stack platform ingesting patient medical histories (PDFs) into structured clinical knowledge wikis using adversarial dual-model iteration (Claude Opus + GPT-4o) until findings converge
- 80% LLM cost reduction via tiered model strategy; compliance-by-design: Fernet encryption, append-only audit log, RBAC, on-device PII anonymization
- Stack: Python, FastAPI, PostgreSQL, React/Next.js, Docker

AlterEco Publishing Pipeline | *Client: alternativaseconomicas.coop*

- Engineered an AI pipeline using Claude Vision API and n8n to automate 40+ articles per issue from PDF to Drupal CMS; reduced per-article processing time from 30 min to under 3 min (10x+ gain)

Soccer Verdun Email Agent | *Client: Soccer Verdun Community Club, Montreal*

- Built a multilingual AI email agent automating responses to player and parent inquiries in EN/FR/ES using RAG over club documents and policies
- Architecture: n8n orchestration, Pinecone vector store, NaN.builders (qwen3.6) for LLM inference, OAuth-authenticated Gmail integration; model swap and OAuth migration handled in production with zero downtime
- Stack: n8n, Pinecone, FastAPI, NaN.builders (qwen3.6), Gmail OAuth

FitTrack | *Personal Project | AI-Powered Mobile Fitness App*

- Built a full-stack mobile fitness tracking app with AI food label scanning (vision model), workout logging, and progress stats
- Stack: Expo SDK 54, React Native, TypeScript, Supabase, NaN.builders (qwen3.6 + mimo-v2.5 for vision)

PROFESSIONAL EXPERIENCE

AI Integration & Legacy Modernization Specialist | Fresche Solutions | Montreal, QC

2014 – Present

- Lead modernization of IBM i legacy systems for enterprise clients across healthcare, manufacturing, and financial services, translating complex 4GL/COBOL workflows into modern Python/SQL architectures
- Proposed and developed internal AI-assisted COBOL modernization pilot leveraging X-Analysis static analysis tool, applying LLM-guided code transformation to legacy codebase
- Technical bridge between legacy business logic and modern stack: scoping, architecture, delivery

Independent Consulting | Self-Employed

2006 – Present

Full-stack delivery, technical leadership, and AI solutions architecture for enterprise clients across financial services, healthcare, and manufacturing.

Full-Stack Developer | Healthcare: Progressive Web Application

2019 – 2021

- Built a mobile-first PWA for on-site nurses with manager approval workflows and offline capability
- Stack: Node.js, React, REST API, PostgreSQL

AI Solutions Architect / Tech Lead | Healthcare: Medical Rehabilitation

2012 – 2015

- Delivered a medical rehabilitation billing system covering patient billing, insurance processing, and financial reporting; automated pipelines reduced manual processing by 60%

Senior Developer / Tech Lead | Manufacturing: Commercial Kitchen Equipment

2006 – 2016

- Designed enterprise ticketing and workflow system; reduced cross-department resolution time by 40%

TECHNICAL SKILLS

AI / ML: LangChain, LangGraph, LangSmith, Agentic Workflows, Multi-agent Systems, LLM Orchestration, RAG, Prompt Engineering, Model Evaluation, n8n, Streamlit, LLM APIs (OpenAI, Anthropic)

Databases & Vector Stores: PostgreSQL, MySQL, MSSQL, Pinecone, ChromaDB, pgvector, Vector Embeddings, Query Optimization, Data Engineering

Backend & APIs: Python, FastAPI, Node.js, PHP, ColdFusion, SQL, RESTful APIs, API Security

Frontend & DevOps: React, Next.js, TypeScript, JavaScript, Docker, Traefik, GitHub Actions, CI/CD, Cloud Infrastructure

Spoken Languages: English (fluent), Spanish (fluent), French (basic)

EDUCATION

Master of Science in Artificial Intelligence (Expected 2026)

2024 – Present

University of Liverpool, England | Dissertation: WorkflowSynth: LLM-guided programme synthesis with formal verification for enterprise workflows

Microcomputer Programming & Web Developer Certificates

1999 – 2001

Centennial College, Toronto, ON

Earlier studies in Electronic Engineering, Sound Design, and Technical Electronics. Buenos Aires, Argentina.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- LangChain Developer Certification | LangChain Inc.
- Agentic RAG Specialization | Coursera
- AI & Automation Specializations | Coursera / Udemy